The simplest computers

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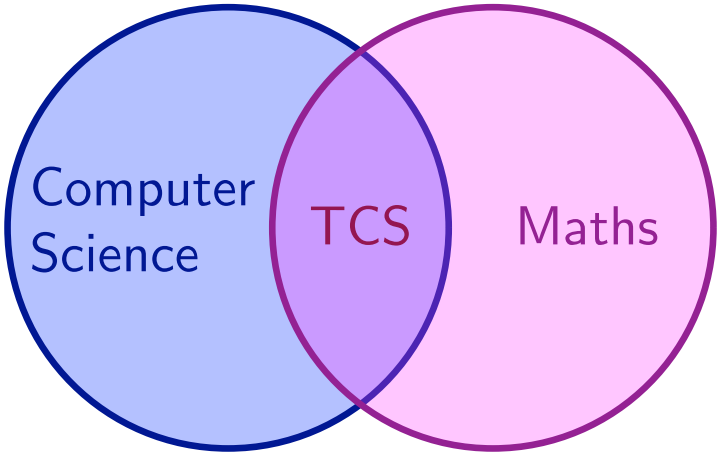

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Models of computers

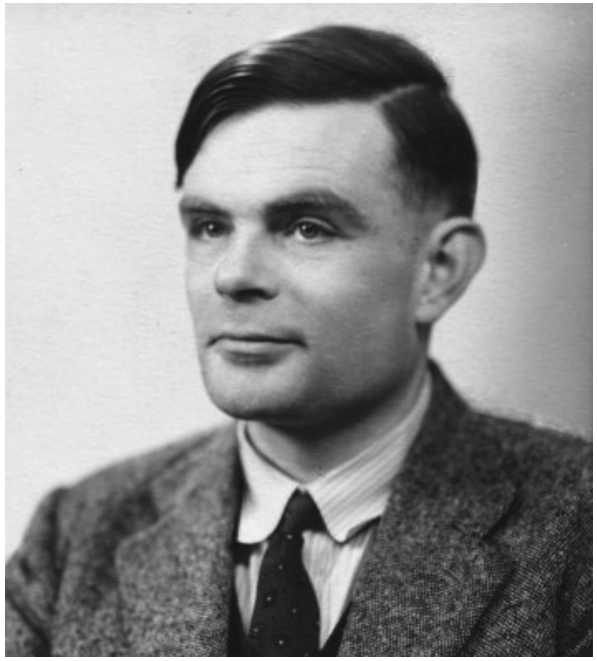
- Real-world computers are too complicated to mathematically analyse.
- Instead, we use mathematical models of computers, which are simple enough to analyse, but powerful enough to tell us something about real-world computers.

A model computer should be:

- complicated enough to carry out any computation we can imagine,
- simple enough to analyse using mathematical ideas.

Turing machines

The most famous example of a model computer is a Turing machine.



Alan Turing
(1912 – 1954)

Church-Turing Thesis

Most useful model computers are Turing-complete. This means that any model computer can simulate any other model computer.

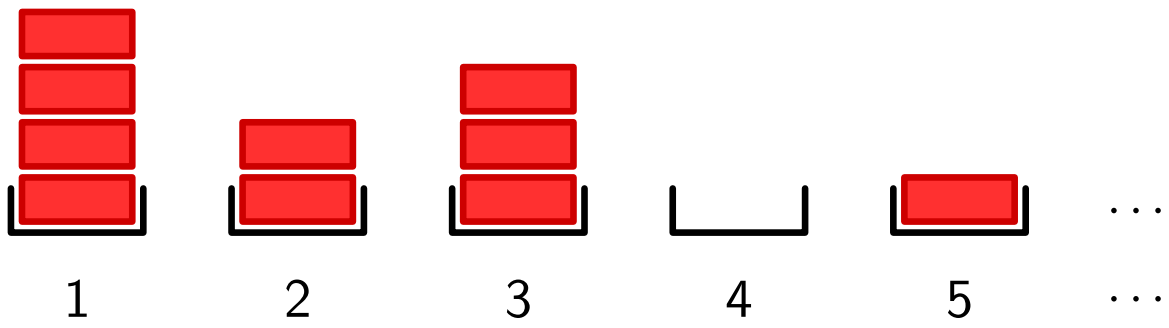
We believe any imaginable program can be simulated by any Turing-complete machine. This is known as the Church-Turing Thesis.

Counter machines

We will study a Turing-complete model computer called a counter machine.

A counter machine consists of a sequence of registers (stacks). Each register contains a number of blocks.

We have a very simple programming language that lets us change the number of the blocks in the registers.



Counter machines

The programming instructions come in seven types:

- `INC` $\#A$ (Increase register $\#A$ by one)
- `DEC` $\#A$ (Decrease register $\#A$ by one — if empty, no change)
- `COPY` $\#A$, $\#B$ (Copy register $\#A$ to $\#B$ — overwrites B, no change to A)
- `CLR` $\#A$ (Clear register $\#A$)
- `GOTO` $\#N$ (Go to the instruction on line $\#N$)
- `TEST` $\#A$, $\#N$ (If register $\#A$ is empty, go to line $\#N$)
- `HALT`

Instructions

- Go to `https://github.com/dillon128/CounterMachines`
- Select `CounterMachines.txt`
- Select the code in this file and copy it using `Ctrl-C`.
- Open `https://sagecell.sagemath.org/` in another tab.
- Paste the text into the cell using `Ctrl-V`.

Challenges

Assume that x, y, and z are the number of blocks in registers #1, #2, and #3.

Program counter machines to:

(1) calculate xy,
(2) calculate 2^x,
(3) calculate x^y,
(4) calculate the absolute value of $x - y$,
(5) sort x and y into increasing order,
(6) find the remainder after dividing x by y,
(7) sort x, y, and z into increasing order,
(8) test to see if x is prime.

The Busy Beaver challenge

The challenge is to write a computer program that will run as long as possible using some fixed number of commands.

Rules:

- The program must eventually halt — no infinite loops!
- You can't give the program a huge amount of initial data to read — all programs start with empty input.

The Busy Beaver challenge

Mathematicians recently solved one case of this problem.

COMPUTABILITY

With Fifth Busy Beaver, Researchers Approach Computation's Limits

8 | *After decades of uncertainty, a motley team of programmers has proved precisely how complicated simple computer programs can get.*

The Fifth Busy Beaver Finally Found: A Triumph of Collaboration and Computation

AS 24 Apr 2025

JULY 25, 2024 | 8 MIN READ

Mathematicians Have Finally Found the Fifth 'Busiest Beaver'

The busy beaver function is unpredictable. But now, after more than 40 years, the fifth value of the function has been revealed

The Busy Beaver problem

The Busy Beaver problem is essentially impossible to solve.

If there was a good way to solve the Busy Beaver problem, then we could find solutions to some of the most famous open problems in mathematics.

The Busy Beaver problem

The Goldbach Conjecture

If n is an even positive integer, and $n > 2$, then n can be expressed as the sum of two primes.

Examples

$4 = 2 + 2$, $6 = 3 + 3$, $8 = 5 + 3$, $10 = 7 + 3$, $12 = 7 + 5$,...

We could construct a counter machine that examines each positive even integer in turn and tests to see if it can be written as the sum of two primes.

If it discovers a counterexample to the Goldbach Conjecture, then it will halt, otherwise it will run forever.

To find if the Goldbach Conjecture is true, we need to decide whether the machine will run forever.

The Busy Beaver problem

Imagine that we can write this program with at most 50 instructions, and we know that the solution to the Busy Beaver problem for 50 instructions is a machine that runs for 50^7 steps.

We could run our program for $50^7 + 1$ steps. If it halts before that point, then we know there is a counterexample to the Goldbach Conjecture.

If it is still running at $50^7 + 1$ steps, then we know it will run forever, or else it would be the solution to the busy beaver problem. Therefore there is no counterexample, so the Goldbach Conjecture is true.

The Busy Beaver problem

My personal records for the Busy Beaver problem.

Number of lines	Number of steps before halting
7	14
12	46
17	

The Busy Beaver problem

My personal records for the Busy Beaver problem.

Number of lines	Number of steps before halting
7	14
12	46
17	17,433,922,066